

Aditya Ganguli

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EDUCATION

McMaster University

Bachelor of Computer Engineering

Cumulative GPA: 3.86/4.0; Dean's List 2024-2025, 2025-2026

Relevant Coursework: Data Structures & Algorithms, Embedded Systems Design, Digital Logic, Computer Architecture

Hamilton, Ontario

Expected May 2029

WORK EXPERIENCE

Zurich Canada Holdings Limited

IT Core Systems IV Intern

Toronto, Ontario

May 2025 – Aug 2025

- Led the design and deployment of a Datadog error monitoring system across multiple PolicyCenter environments, tracking a comprehensive array of runtime errors to enhance observability and incident response.
- Engineered an Excel-VBA data pipeline to aggregate, sort, and analyze 14,000+ Guidewire policies, generating structured summary reports across multiple datasets.
- Engineered data-driven Jira dashboards for Commercial Auto and Middle Market teams to track sprint efficiency and bottlenecks, improving delivery and decision-making.

PROJECTS

GBU - Gameboy Emulator written in C++

May 2026

- Engineered a low-level Game Boy hardware emulator from scratch in C++, utilizing SDL2 to handle real-time graphics rendering, window management, and peripheral input mapping.
- Simulated the internal computer architecture of the Sharp LR35902 CPU, implementing cycle-accurate opcode decoding, register files, and an integrated Memory Management Unit (MMU).
- Developed memory bank switching (MBC) controllers and persistent cartridge save-state logic to accurately dump and load SRAM data for AAA titles like Pokémon and Super Mario Land.

Small-Scale LiDAR Scanner

Apr 2026

- Developed a portable LiDAR spatial scanner using a Texas Instruments MSP432 microcontroller, writing the core system control firmware in C within Keil uVision.
- Integrated I2C communication at 100 kbps to interface with a Time-of-Flight distance sensor and implemented a 115200 bps UART link to stream real-time data to a host PC.
- Programmed a stepper motor to execute precise 360-degree, 64-point indoor boundary scans, utilizing a MATLAB script to parse and convert polar coordinates into volumetric 3D wireframe models.

IMPACT Initiative - Assistive Smartphone Case

Apr 2025

- Collaborated within a team to design and 3D-print an ergonomic, accessible assistive smartphone case tailored for a client living with Usher Syndrome.
- Integrated mechanical external shortcut buttons into the case, mapping essential device functions such as magnifiers and hearing aid apps to tactile physical inputs.
- Validated the prototype's efficiency with the Engineering Design Cycle by conducting multi-day field tests, analyzing comfort, ease of use and durability.

VOLUNTEERING

Mississauga Taekwondo

Volunteer Instructor

Mississauga, Ontario

Jul 2020 – Aug 2023

- Dedicated 3+ years to instructing students in martial arts techniques while mentoring youth to develop discipline, structural focus, and leadership skills.

Ontario Konkani Association

Youth Coordinator Member

Mississauga, Ontario

May 2022 – May 2024

- Coordinated youth activities for cultural and community events, including festivals, picnics, and charity initiatives.

ADDITIONAL

Technical Skills: Advanced in Quartus, Autodesk Inventor/AutoCAD, Git, KiCad, SolidWorks, Fusion 360, Arduino, Soldering, ARM Based Microcontrollers (MSP432E401Y, ESP32)

Programming Languages: Advanced in C/C++, Java, JavaScript, Python, Verilog, VHDL, MATLAB, HTML, CSS